IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Data collection: SpaceX REST API (JSON snapshot) and Wikipedia web scraping
- Data wrangling: Falcon 9 filter, PayloadMass imputation, landing Class label, one-hot encoding
- EDA with SQL (SQLite) and with visualization (seaborn)
- Interactive analytics: Folium map and Plotly Dash dashboard
- Predictive analysis: Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV

Summary of all results

- 90 Falcon 9 launches analysed; 60 landed (66.7%)
- Landing success rose from 0% (up to 2013) to 84–90% in 2019–2020
- KSC LC-39A and VAFB SLC-4E landed about 77% of launches vs 60% at CCAFS SLC-40
- All four tuned classifiers tie at 83.3% test accuracy (15 of 18 correct)

Introduction

Project background and context

SpaceX advertises Falcon 9 launches at about $62 million; other providers charge $165 million or more, largely because SpaceX reuses the first stage.

If we can predict whether the first stage will land, we can estimate the cost of a launch, which is useful for a competitor bidding against SpaceX.

Goal: build a machine learning pipeline that predicts first-stage landing success.

Problems to answer

- Which factors determine whether the rocket lands successfully?
- How do the features interact to determine the landing success rate?
- Which operating conditions must be in place to ensure a successful landing program?

Section 1

Methodology

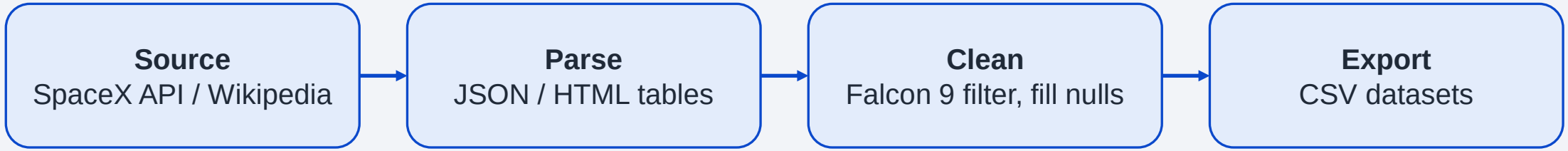


Methodology

- **Data collection**
  - SpaceX REST API (static JSON snapshot after the live API timed out) and Wikipedia web scraping
- **Data wrangling**
  - Falcon 9 filter, mean-imputed PayloadMass, landing Class label (1 = landed, 0 = not), one-hot encoding
- **Exploratory data analysis (EDA) with SQL and data visualization**
- **Interactive visual analytics with Folium and Plotly Dash**
- **Predictive analysis with classification models**
  - Standardize features, 80/20 split, tune with GridSearchCV (10-fold), compare test accuracy

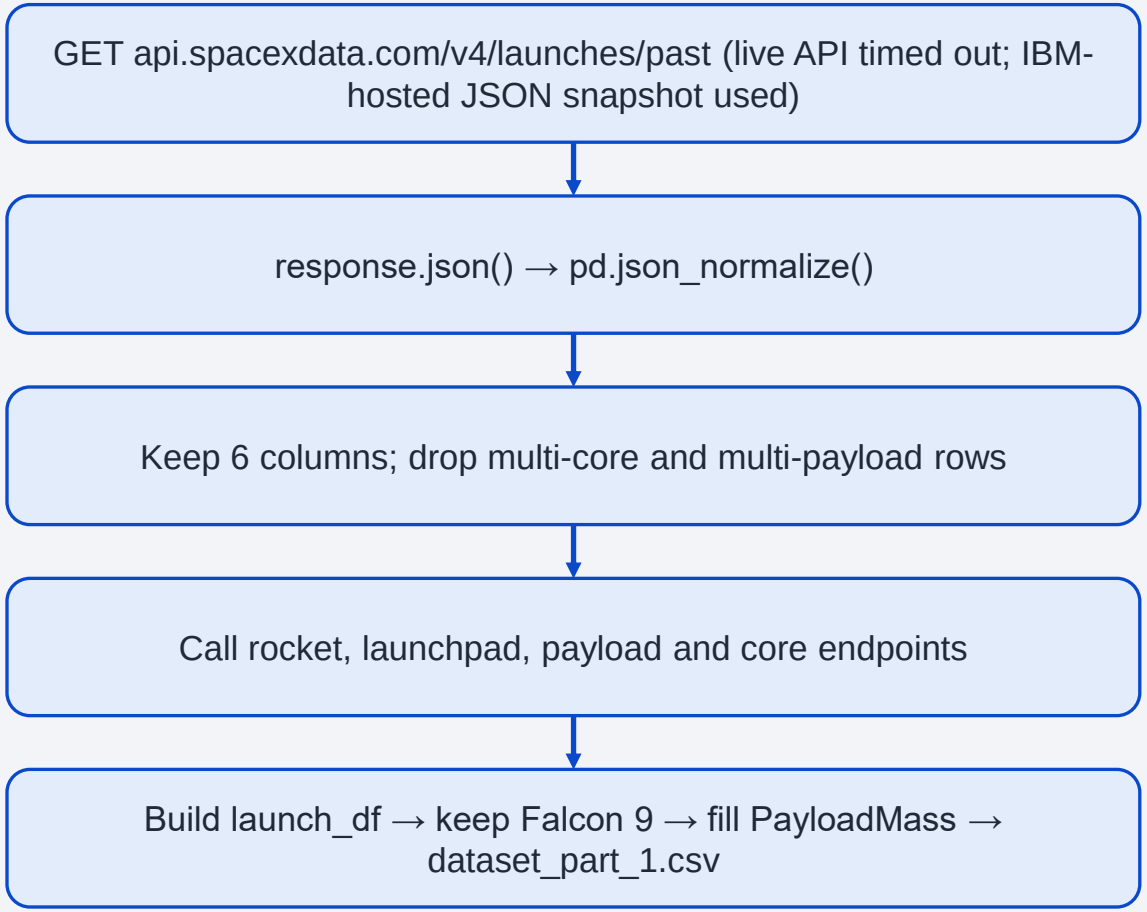
Data Collection

- Two sources: the SpaceX REST API (launch records plus rocket, launchpad, payload and core ID lookups) and the Wikipedia Falcon 9 launch table
- API: requests.get() → .json() → pd.json_normalize(), keeping rocket, payloads, launchpad, cores, flight_number and date
- Helper functions added booster version, launch site and coordinates, payload mass and orbit, block, reuse count and serial
- Filtered to Falcon 9 (90 launches); 5 missing PayloadMass values filled with the mean (6,123.5 kg)
- Web scraping: requests + BeautifulSoup → HTML table → pandas DataFrame (121 rows) → CSV



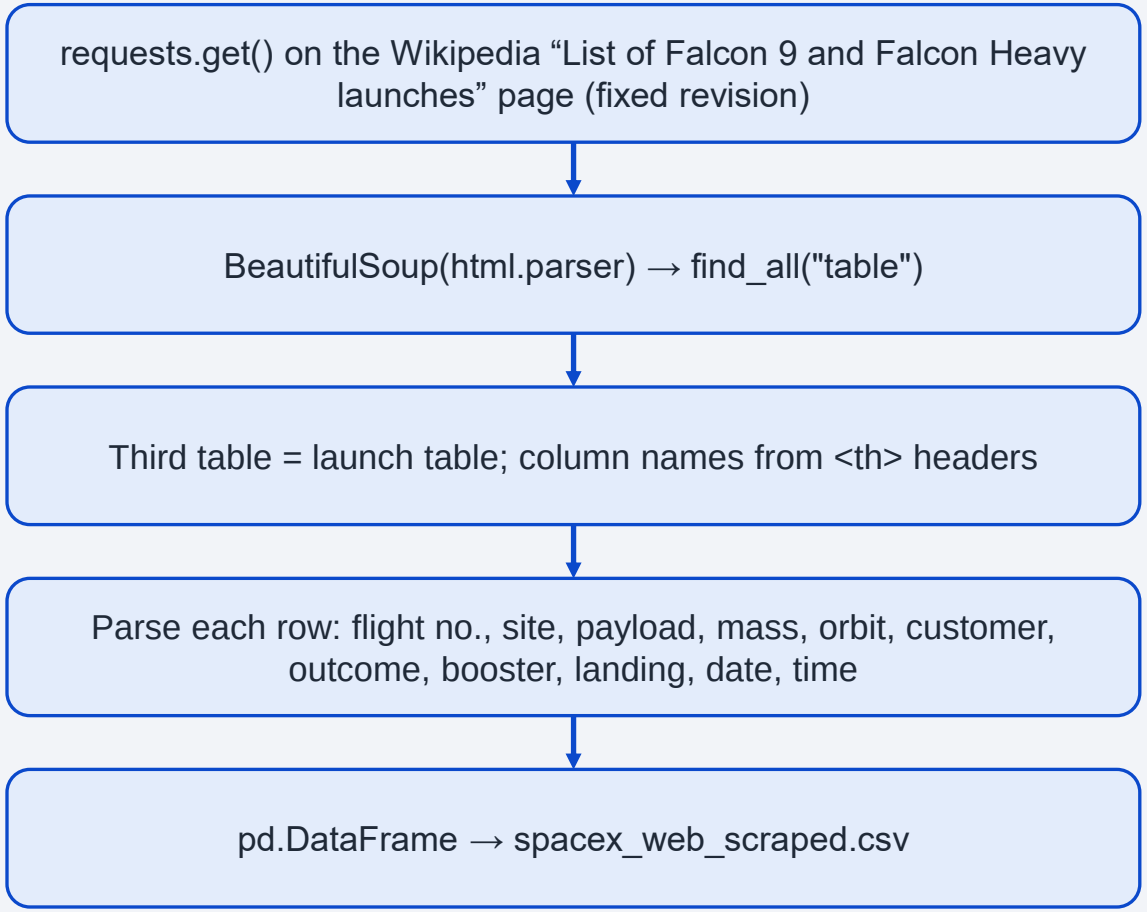
Data Collection – SpaceX API

- Collected launch data with REST GET requests to the SpaceX API
- Decoded the response with .json() and flattened it with json_normalize()
- The live API returned HTTP 522, so a saved JSON snapshot of the same response was used
- Result: 90 Falcon 9 launches; LandingPad stays null for 26 launches
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/01_data_collection_api.ipynb



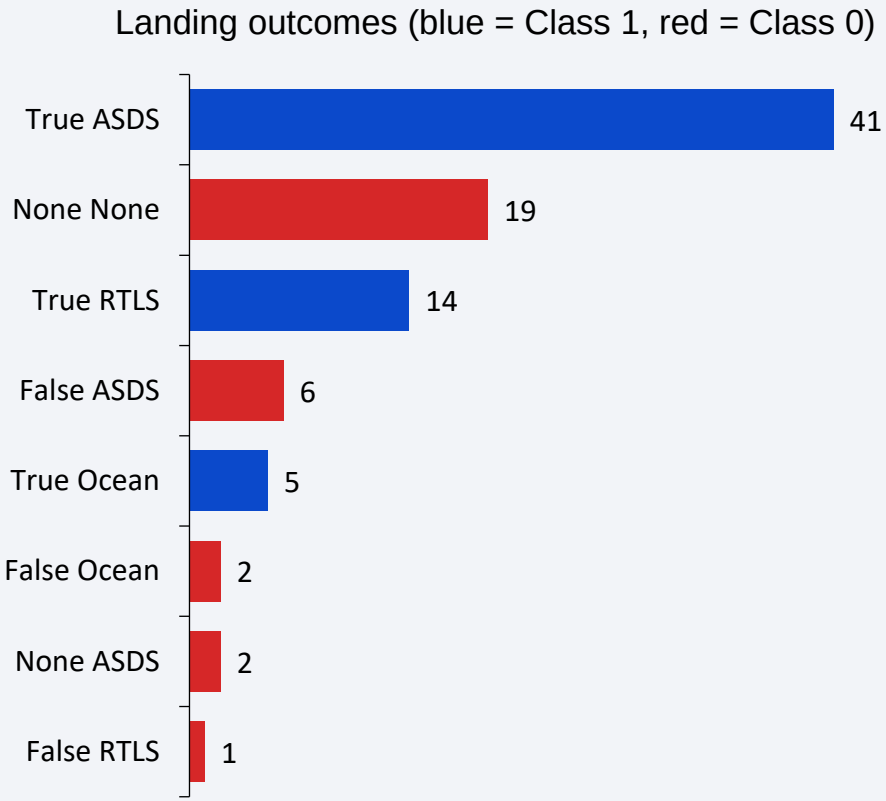
Data Collection - Scraping

- Scraped the Falcon 9 launch records from Wikipedia with BeautifulSoup
- Parsed the launch table row by row into a dictionary, then a pandas DataFrame
- Result: 121 rows × 11 columns saved as spacex_web_scraped.csv
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/02_web_scraping.ipynb



Data Wrangling

- Counted launches per site (CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13) and per orbit (GTO 27, ISS 21, VLEO 14, …)
- Mapped the Outcome column to a Class label: 1 = landed (60), 0 = did not land (30)
- Overall landing success rate: 66.7%
- One-hot encoded Orbit, LaunchSite, LandingPad and Serial → 80 features (dataset_part_3.csv)
- [Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/03_data_wrangling.ipynb](https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/03_data_wrangling.ipynb)



EDA with Data Visualization

- Scatter plots (seaborn catplot, coloured by Class): Flight Number vs Launch Site, Payload vs Launch Site, Flight Number vs Orbit, Payload vs Orbit – to see how landing outcome depends on experience, payload and destination
- Bar chart: success rate of each orbit type – to compare orbits on one scale
- Line chart: average success rate by year – to show the learning trend over time
- Feature engineering: selected 12 candidate features and one-hot encoded the categorical ones for modelling
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/05_eda_data_visualization.ipynb

EDA with SQL

- Loaded Spacex.csv (101 rows) into SQLite (my_data1.db) and queried it from Jupyter with ipython-sql
- Unique launch sites; five records for sites starting with “CCA” (LIKE, LIMIT)
- Total payload mass for NASA (CRS): SUM; average payload mass for booster F9 v1.1: AVG
- First successful ground-pad landing date: MIN(Date)
- Drone-ship successes with payload between 4,000 and 6,000 kg (WHERE … AND)
- Successful vs failed mission outcomes (GROUP BY); boosters with maximum payload (subquery with MAX)
- 2015 drone-ship landing failures (month, booster, site); landing outcomes ranked between 2010-06-04 and 2017-03-20 (GROUP BY, ORDER BY)
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/04_eda_sql_analysis.ipynb

Build an Interactive Map with Folium

- Launch sites: a circle (1,000 m radius) with a popup and a text label at each site – shows where the sites are
- Marker cluster of every launch: green marker = landed (class 1), red marker = not landed (class 0) – shows which sites perform best
- MousePosition plugin to read coordinates, and a haversine distance function (R = 6,373 km)
- Marker plus PolyLine from CCAFS SLC-40 to the nearest coastline point (0.51 km) and to Melbourne, FL (53.4 km), each labelled with its distance – shows the site is near the coast but far from cities
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/06_launch_site_location_analysis.ipynb

Build a Dashboard with Plotly Dash

- Dropdown: choose All Sites or one launch site – filters every chart
- Pie chart: successful launches by site (All Sites) or success vs failure for the selected site
- Range slider (0–10,000 kg, step 1,000): restricts the payload range
- Scatter chart: payload mass vs launch outcome, coloured by booster version category
- Two callbacks connect the dropdown and slider to the pie and scatter charts so the user can explore which payloads and boosters succeed
- Code: spacex-dash-app.py – Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX

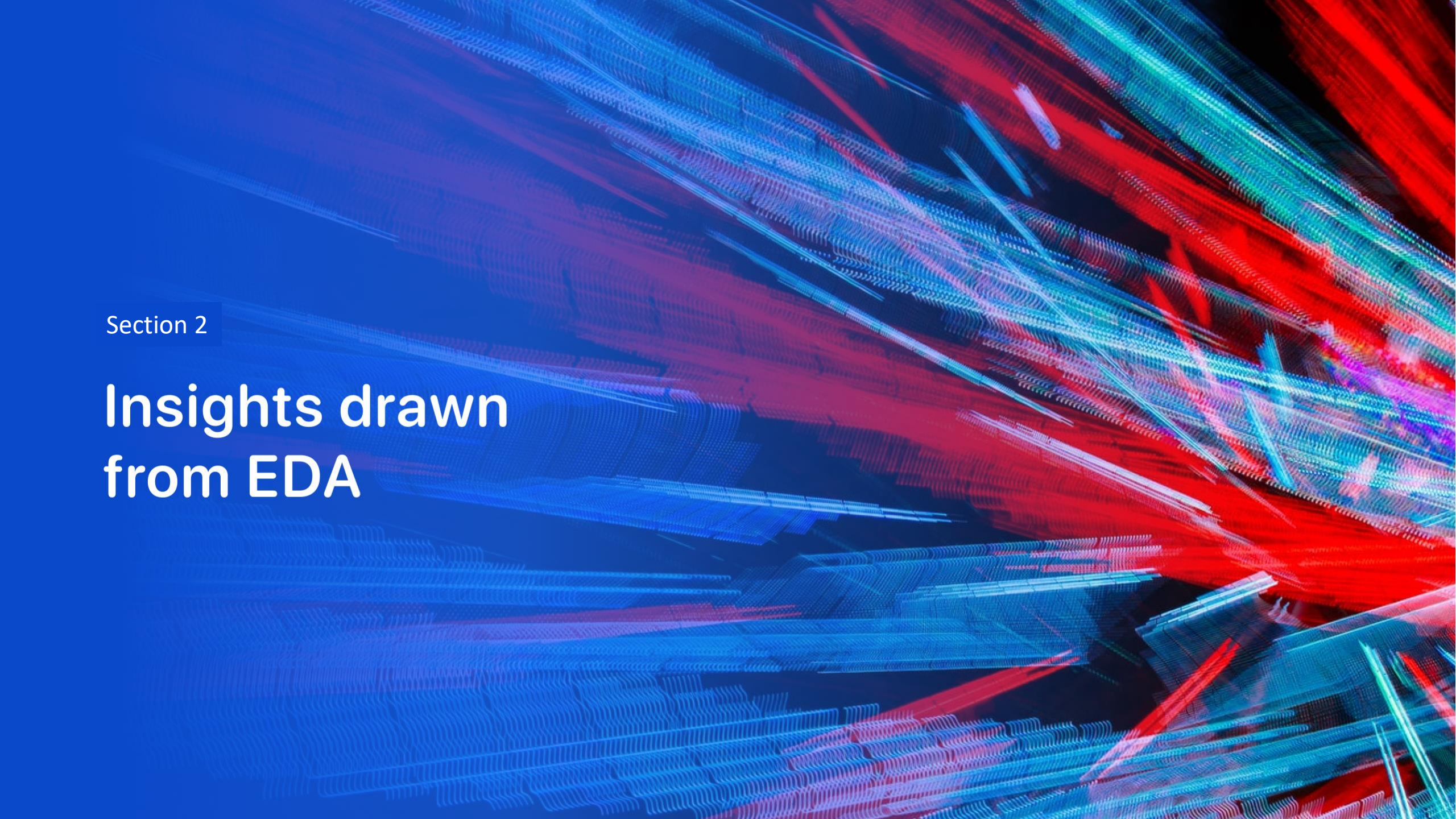
Predictive Analysis (Classification)

- Loaded Class (target) and 80 one-hot features; standardized them with StandardScaler
- Split 80/20 (random_state = 2): 72 training and 18 test launches
- Tuned Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV (10-fold cross-validation)
- Evaluated with test accuracy and confusion matrices, and compared best cross-validation scores
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/07_machine_learning_prediction.ipynb



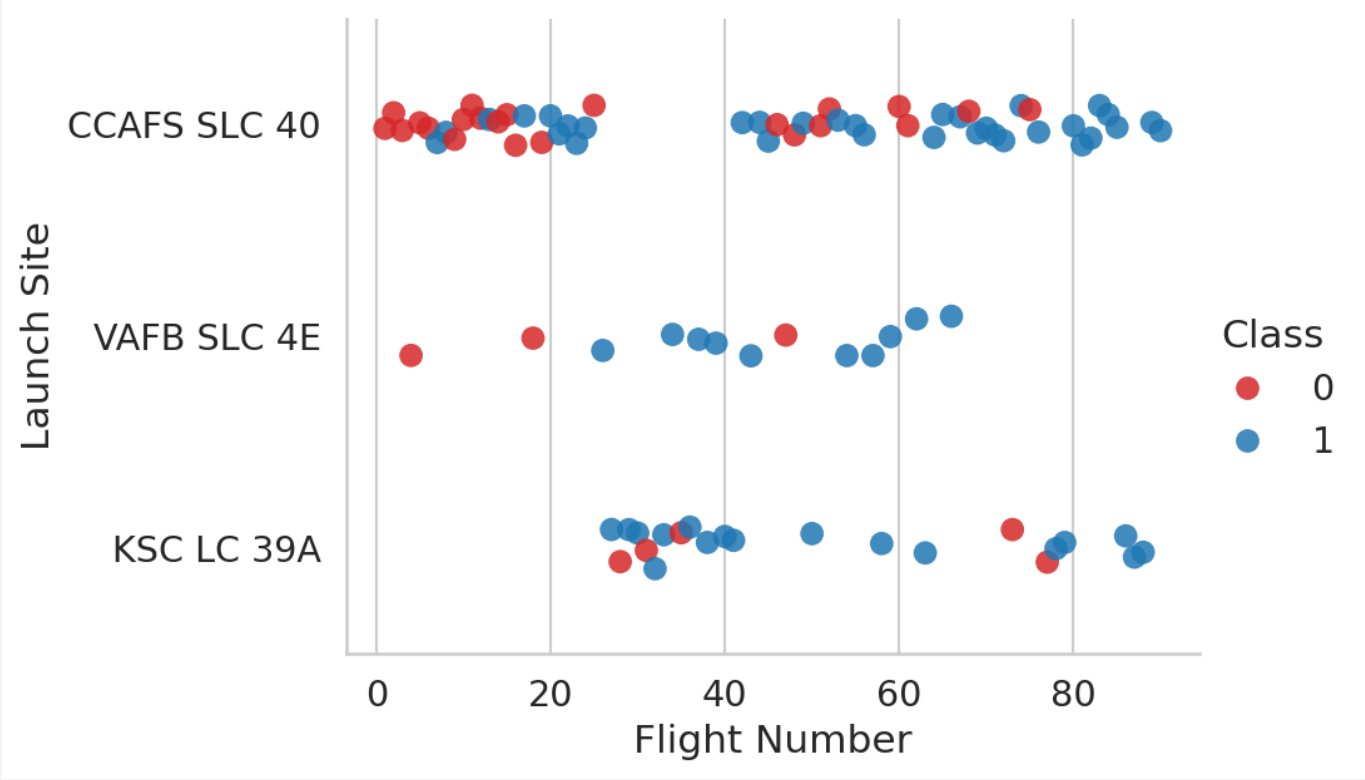
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results


Section 2
Insights drawn from EDA

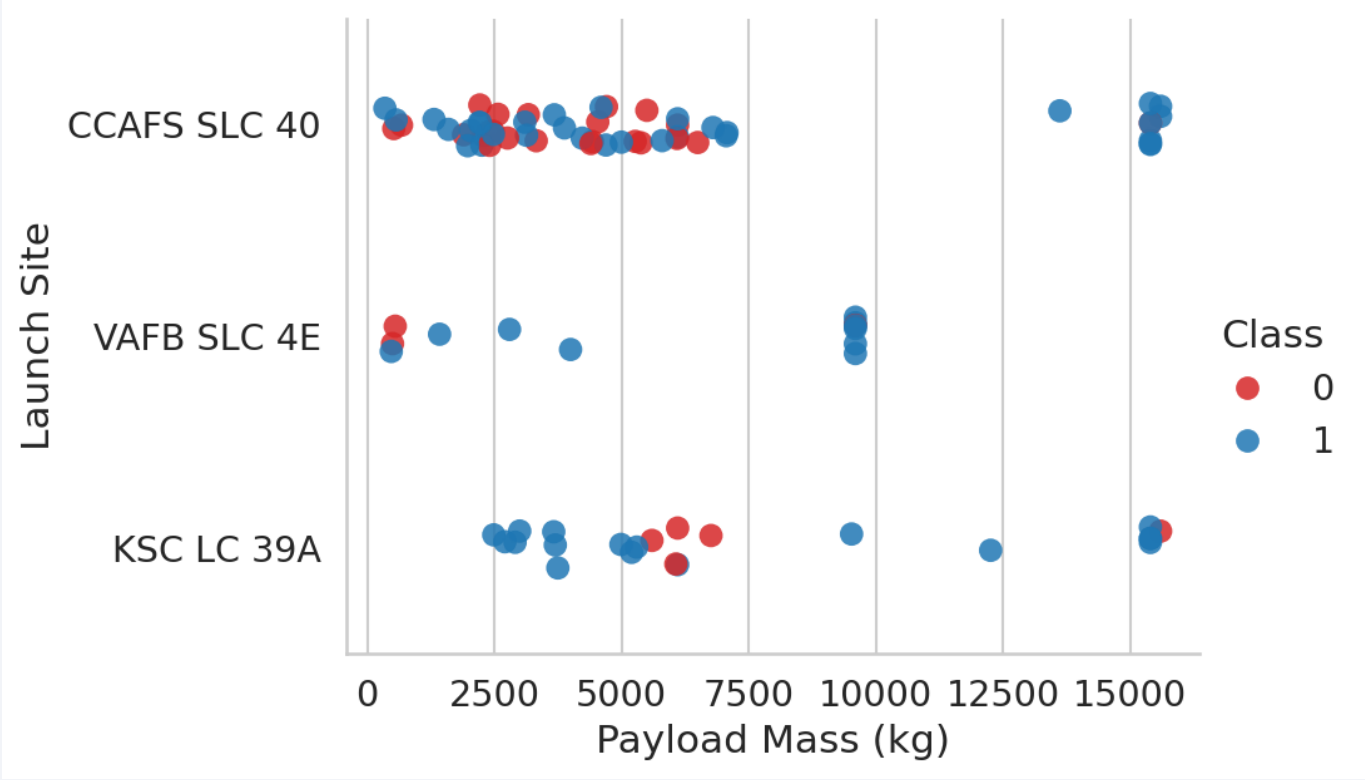
Flight Number vs. Launch Site

- Success rate rises with flight number: 43% for flights 1–30, 73% for 31–60, 83% for 61–90
- CCAFS SLC 40 hosts most launches (55); its early flights mostly failed (red), later ones mostly landed (blue)



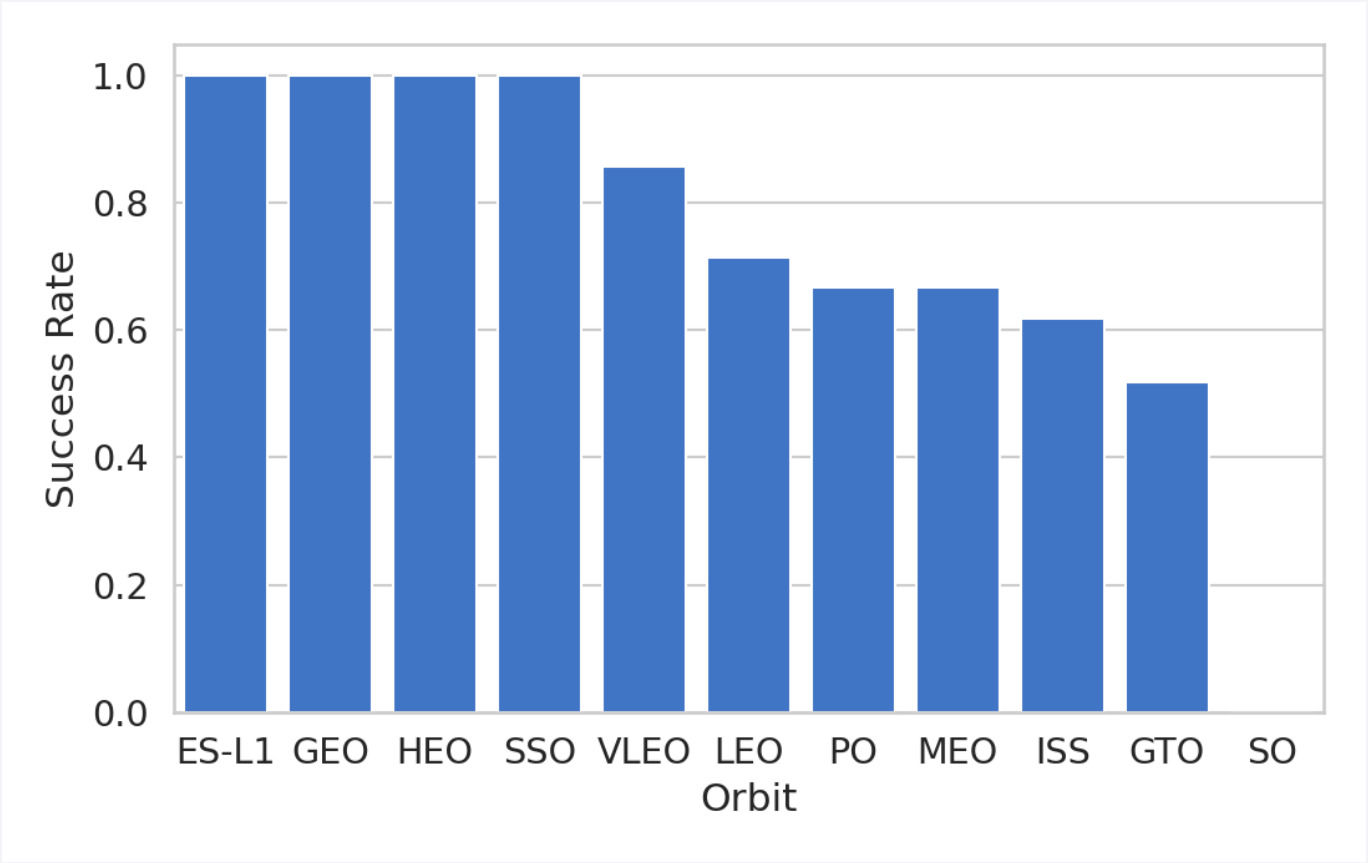
Payload vs. Launch Site

- Payloads above 7,000 kg landed more often at every site: CCAFS SLC 40 91% vs 52%, KSC LC 39A 86% vs 73%, VAFB SLC 4E 86% vs 67%
- Heavy-payload samples are small (11, 7 and 7 launches), so treat this as a trend



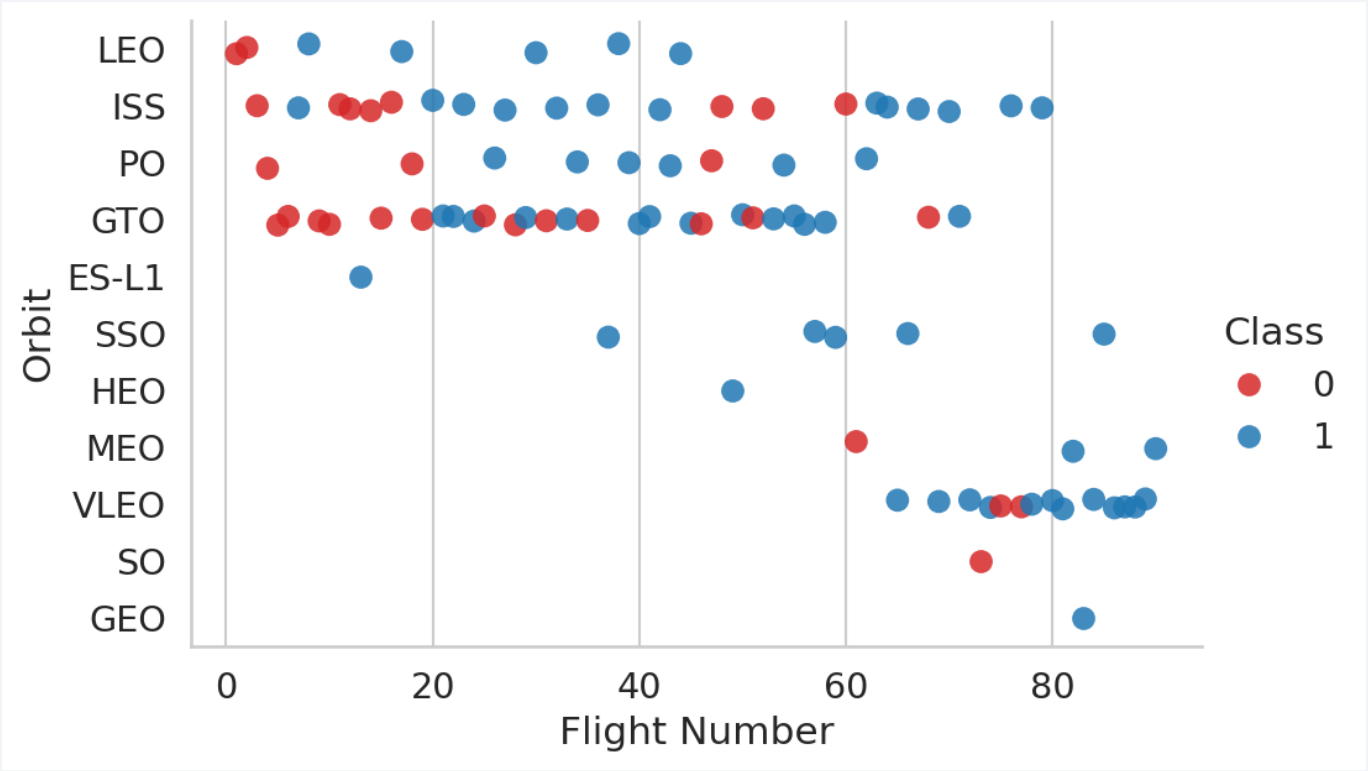
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO reached 100%, but with only 1 to 5 launches each
- VLEO 86%, LEO 71%, PO and MEO 67%, ISS 62%, GTO 52%; SO 0% (one launch)



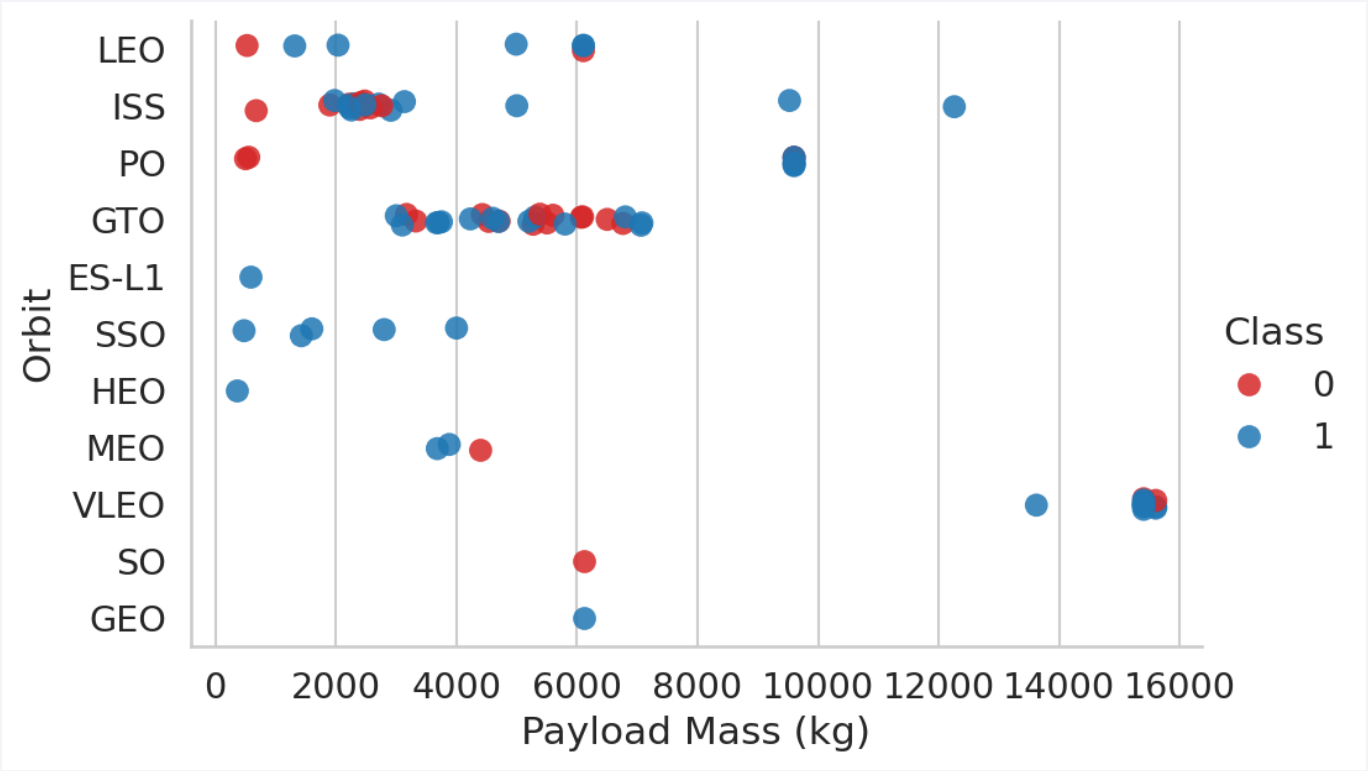
Flight Number vs. Orbit Type

- LEO launches all happened early (flights 1–44); VLEO launches are all recent (flights 65–89) and 86% landed
- GTO spans flights 5–71 with mixed results (52% landed)



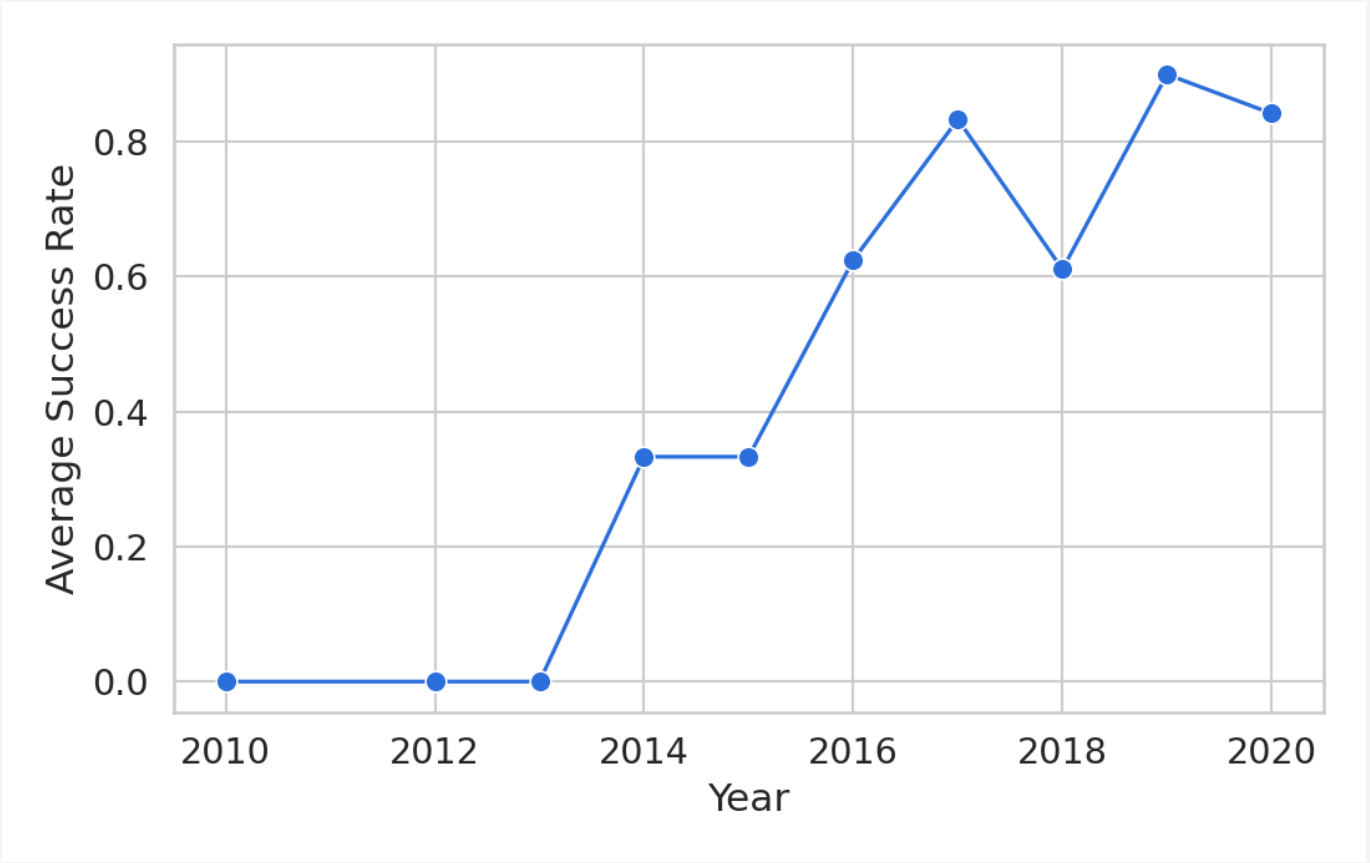
Payload vs. Orbit Type

- VLEO carries the heaviest payloads (13,620–15,600 kg) and landed 86% of the time
- For LEO, ISS and PO, payloads above 6,000 kg landed 83% (12 launches) vs 50% below 3,000 kg (22 launches)
- GTO payloads are 3,000–7,076 kg with 52% success



Launch Success Yearly Trend

- Success was 0% up to 2013, 33% in 2014–2015, 63% in 2016 and 83% in 2017
- A dip to 61% in 2018 was followed by 90% in 2019 and 84% in 2020



All Launch Site Names

Find the names of the unique launch sites

```
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- DISTINCT returns each launch site once: four sites in total (three in Florida, one in California)

Launch Site Names Begin with 'CCA'

Find 5 records where launch sites begin with “CCA”

```
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster	Launch site	Payload (kg)	Orbit	Customer	Landing outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0	LEO	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0	LEO (ISS)	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525	LEO (ISS)	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500	LEO (ISS)	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677	LEO (ISS)	NASA (CRS)	No attempt

- LIKE 'CCA%' keeps launch sites that begin with “CCA”; LIMIT 5 returns the first five records.

Total Payload Mass

Calculate the total payload carried by boosters from NASA (CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass_KG
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

45,596 kg

- SUM adds up the payload mass of all NASA (CRS) launches.

Average Payload Mass by F9 v1.1

Calculate the average payload mass carried by booster version F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass_KG
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

2,928.4 kg

- AVG over the F9 v1.1 launches gives an average payload of about 2.9 tonnes.

First Successful Ground Landing Date

Find the date of the first successful landing outcome on a ground pad

```
SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing
FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

2015-12-22

- MIN(Date) over successful ground-pad landings: the first one was on 22 December 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

List boosters that landed successfully on a drone ship with payload between 4,000 and 6,000 kg

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- WHERE filters drone-ship successes and AND restricts the payload to more than 4,000 kg and less than 6,000 kg: four boosters qualify.

Total Number of Successful and Failure Mission Outcomes

Calculate the total number of successful and failure mission outcomes

```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success (stored with trailing space)	1
Success (payload status unclear)	1

- One “Success” value was stored with a trailing space, so it forms its own row.
- Together: 100 successful missions (including one with unclear payload status) and 1 failure (in flight).

Boosters Carried Maximum Payload

List the boosters that carried the maximum payload mass

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3

Booster_Version
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

- The subquery finds the maximum payload (15,600 kg); 12 F9 B5 boosters carried it.

2015 Launch Records

List failed drone-ship landings in 2015 with booster version and launch site

```
SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)'
AND substr(Date,0,5) = '2015';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Two drone-ship landing attempts failed in 2015 (January and April), both with F9 v1.1 boosters from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank landing outcomes between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- GROUP BY counts each landing outcome in the date range and ORDER BY … DESC ranks them.
- “No attempt” is most common (10); drone-ship successes and failures are tied at 5 each.


Section 3
Launch Sites
Proximities Analysis

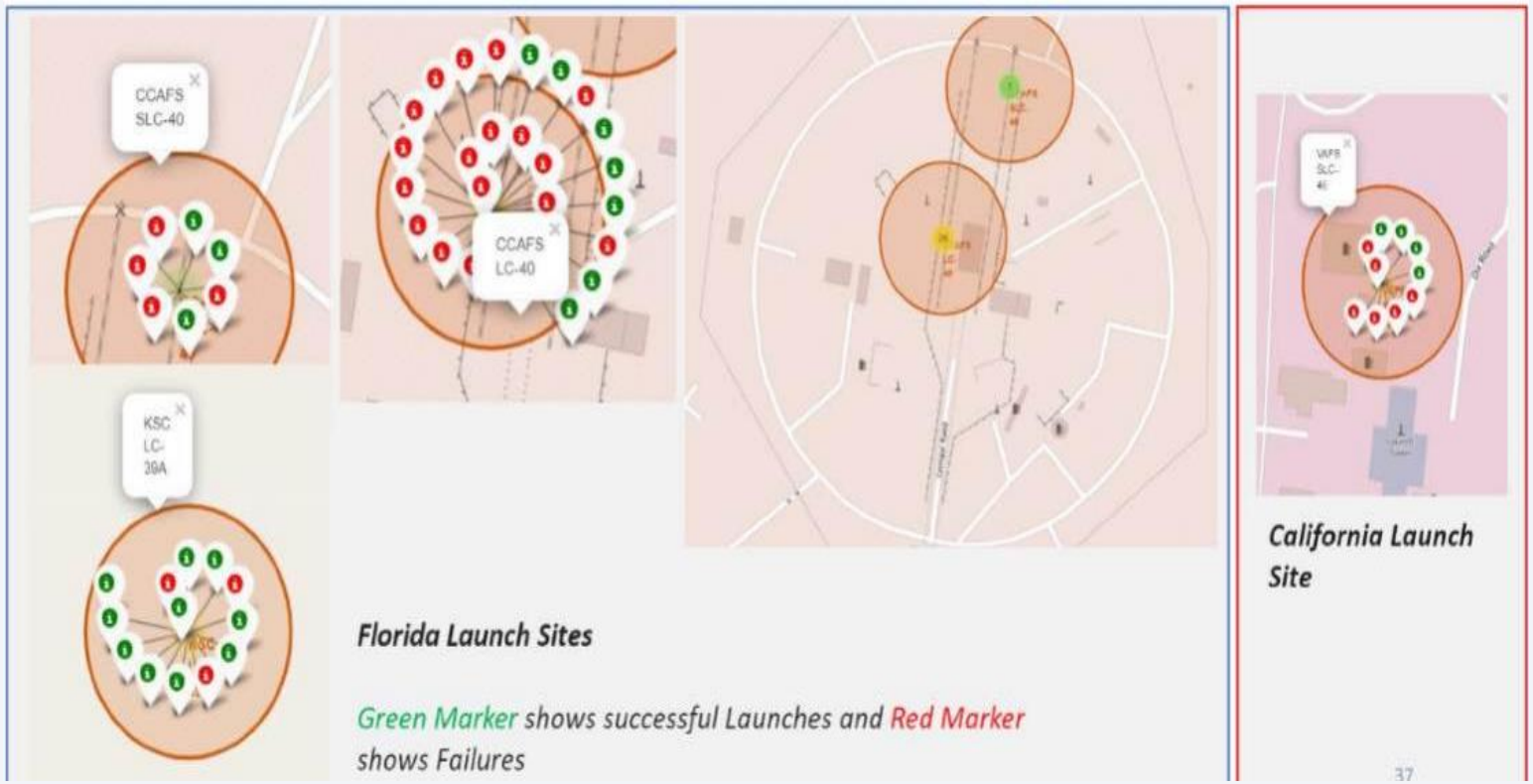
All Launch Sites on a Global Map



- All SpaceX launch sites are on the US coast: Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and California (VAFB SLC-4E)
- Each site is marked with a circle and a text label

Screenshot source: Winning Space Race with Data Science (reference deck)

Launch Outcomes Marked by Color



- Marker clusters group the launches of each site
- Green marker = landed (class 1); red marker = not landed (class 0)
- KSC LC-39A shows mostly green markers, while CCAFS sites show more red
- Florida sites (left and centre) vs the California site VAFB SLC-4E (right)

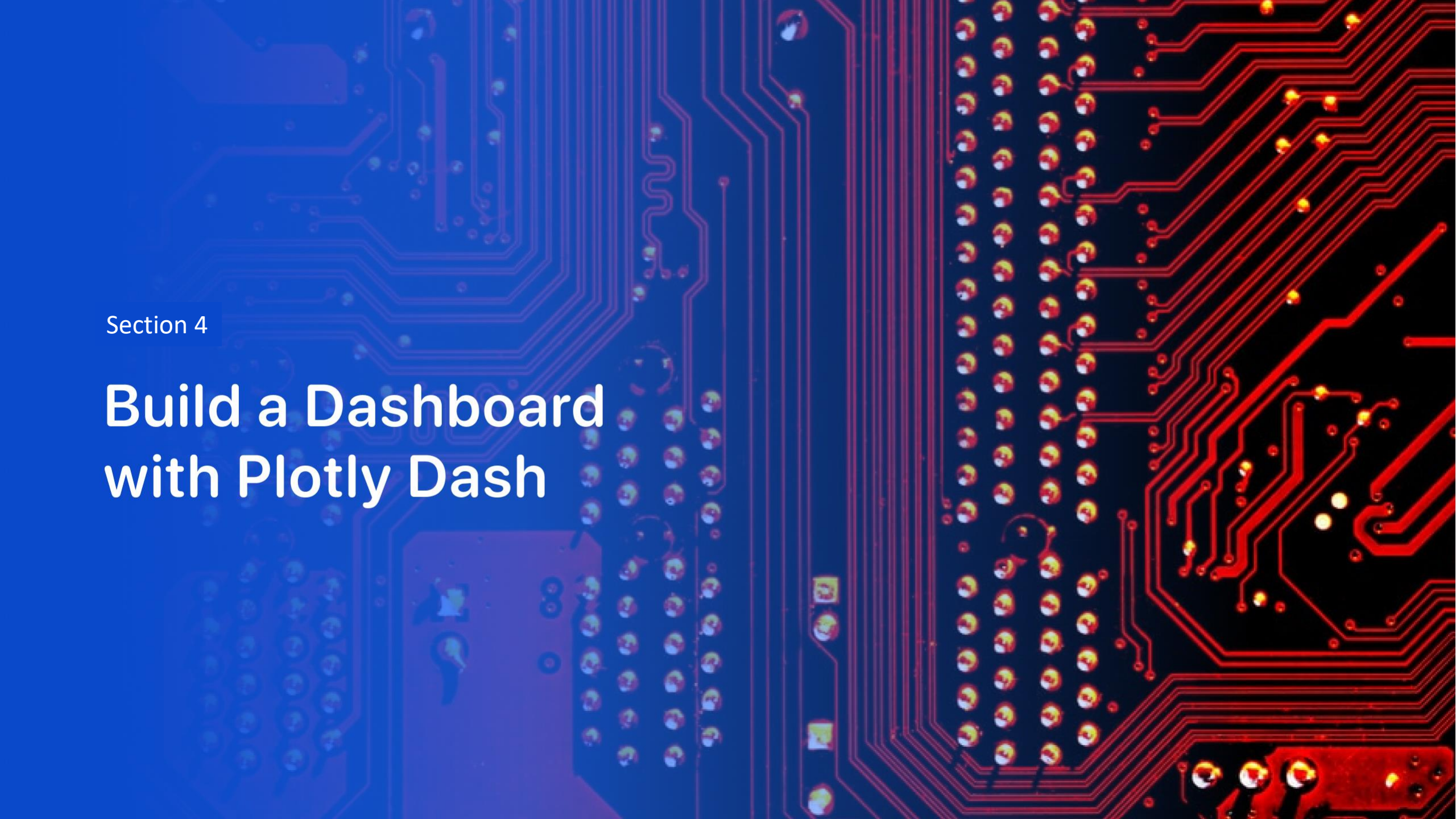
Screenshot source: Winning Space Race with Data Science (reference deck)

Launch Site Proximity: Coast, Highway, Railway and City

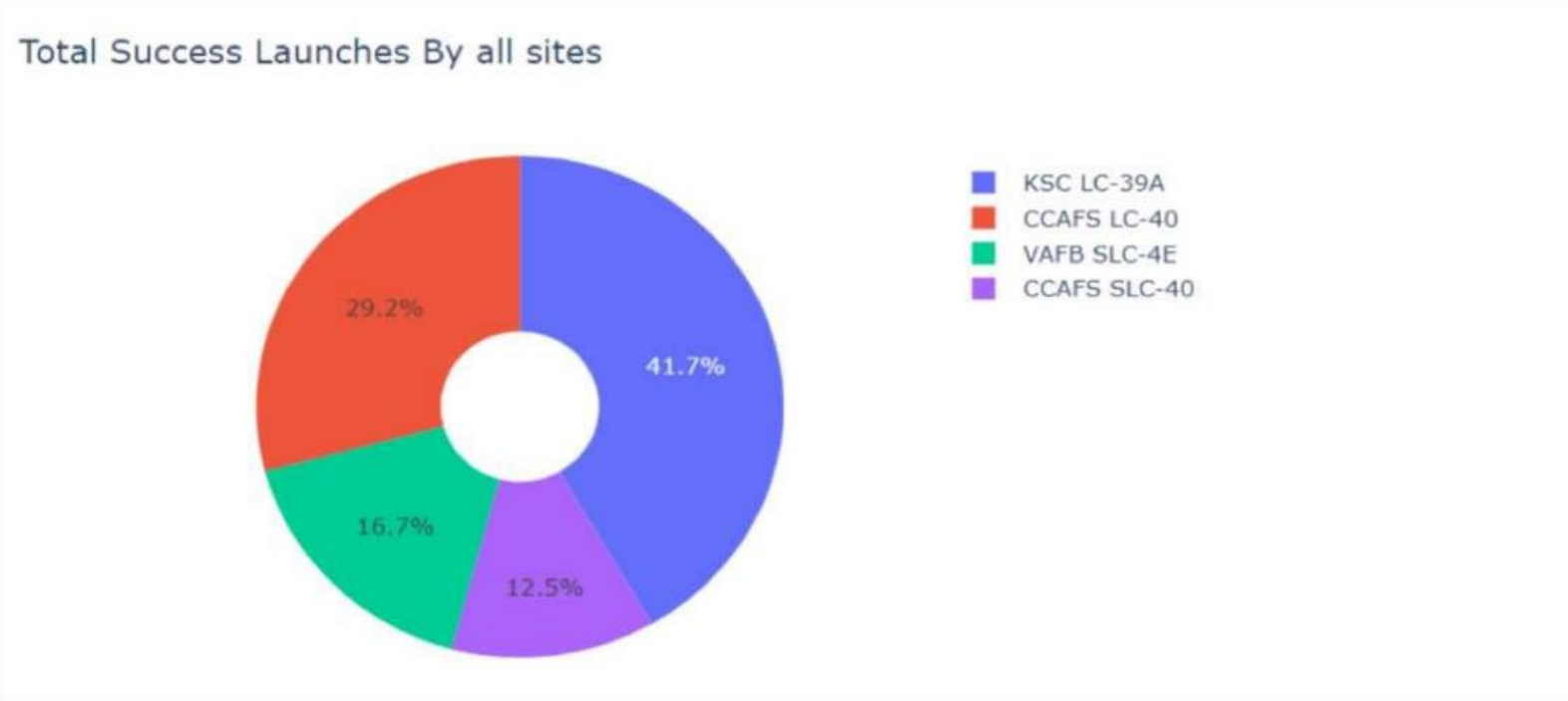


- Distances are drawn as lines with labels (haversine formula)
- My notebook, from CCAFS SLC-40: nearest coastline 0.51 km; Melbourne, FL 53.4 km
- Launch sites sit close to the coast and far from cities, so launches and failed stages fall over water

Screenshot source: reference deck; distance labels in the image come from that run


Section 4
Build a Dashboard with Plotly Dash

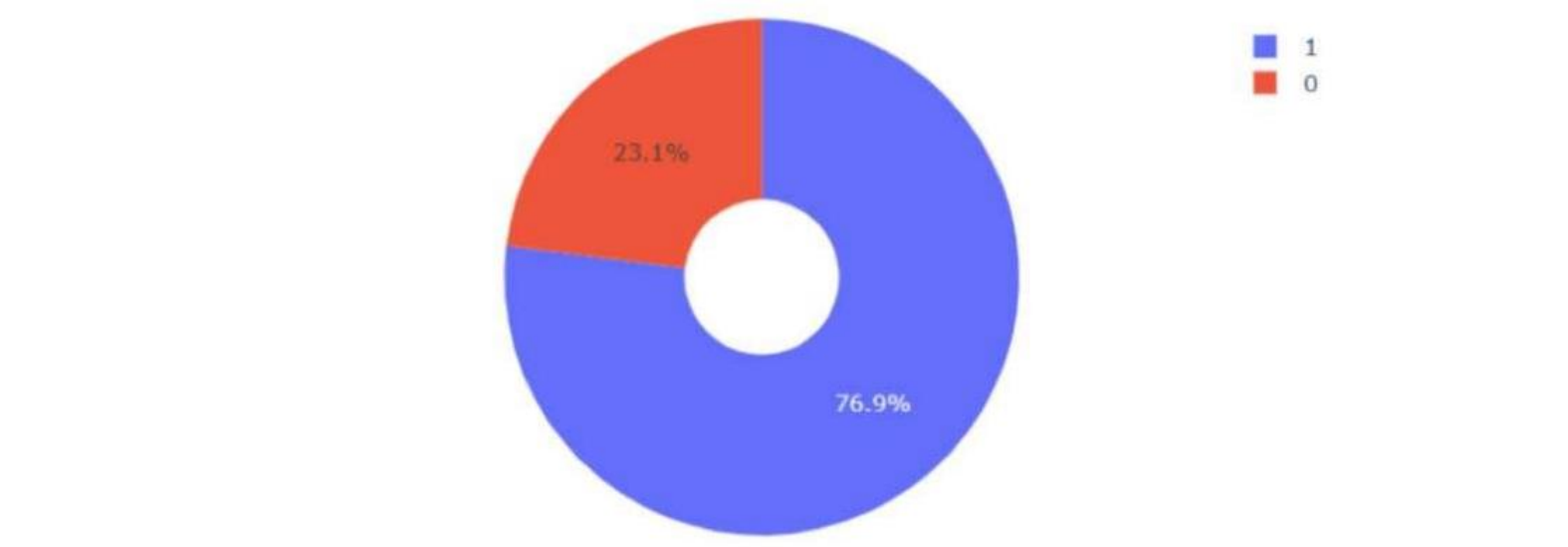
Launch Success Count by Site



- Pie chart for All Sites: share of successful launches per site
- KSC LC-39A has the largest share (41.7%), then CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%)

Screenshot source: Winning Space Race with Data Science (reference deck)

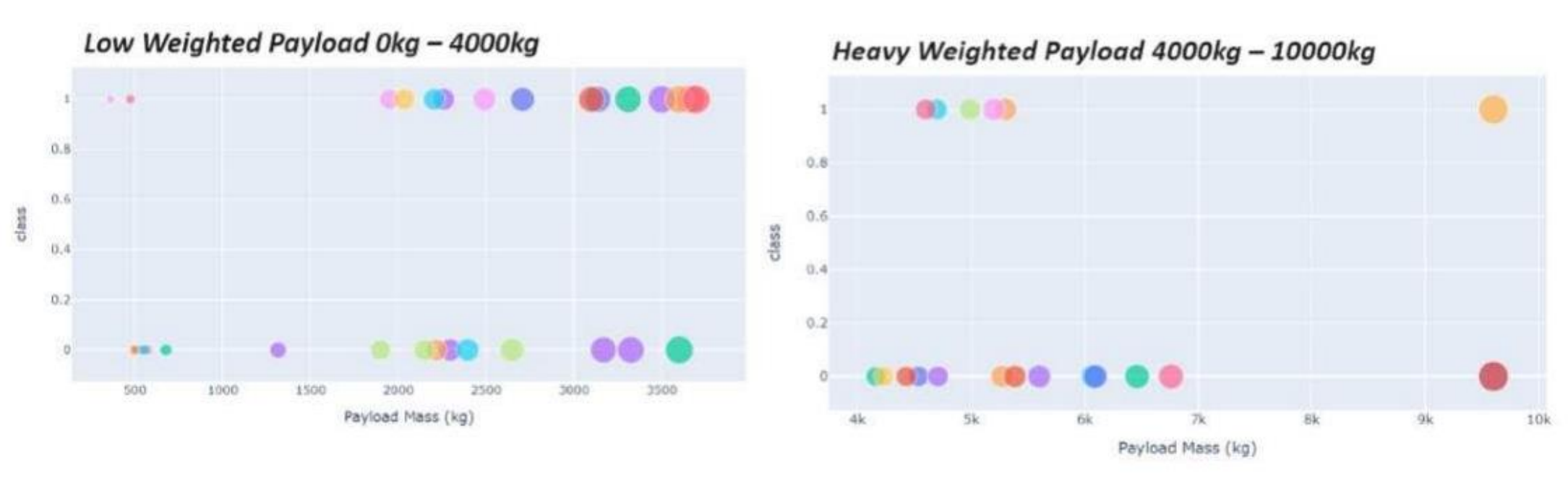
Site with the Highest Success Ratio: KSC LC-39A



- KSC LC-39A has the highest launch success ratio
- 76.9% of its launches succeeded (class 1) and 23.1% failed (class 0)

Screenshot source: Winning Space Race with Data Science (reference deck)

Payload vs. Launch Outcome with Range Slider



- Low payloads (0–4,000 kg, left) show a higher success rate than heavy payloads (4,000–10,000 kg, right)
- Colors are booster version categories; the slider and site dropdown filter the chart

Screenshot source: Winning Space Race with Data Science (reference deck)

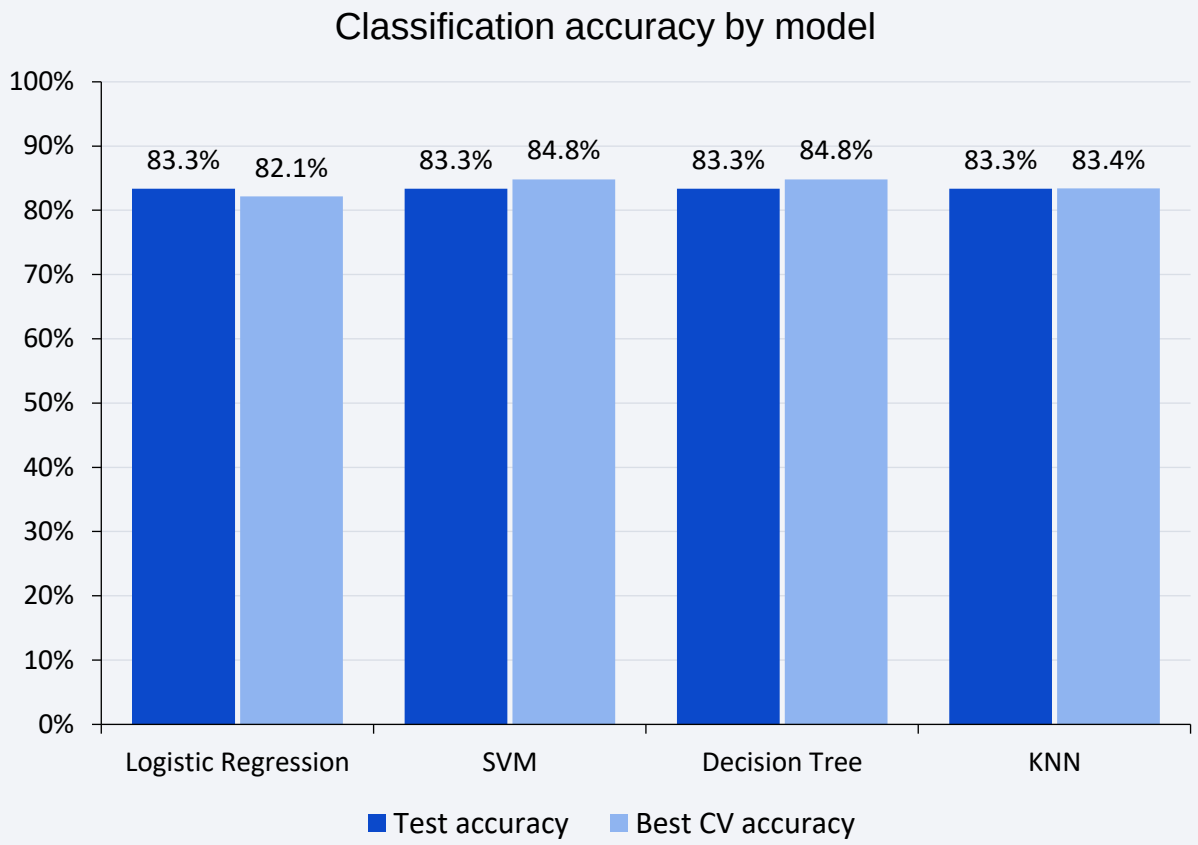
Section 5

Predictive Analysis (Classification)

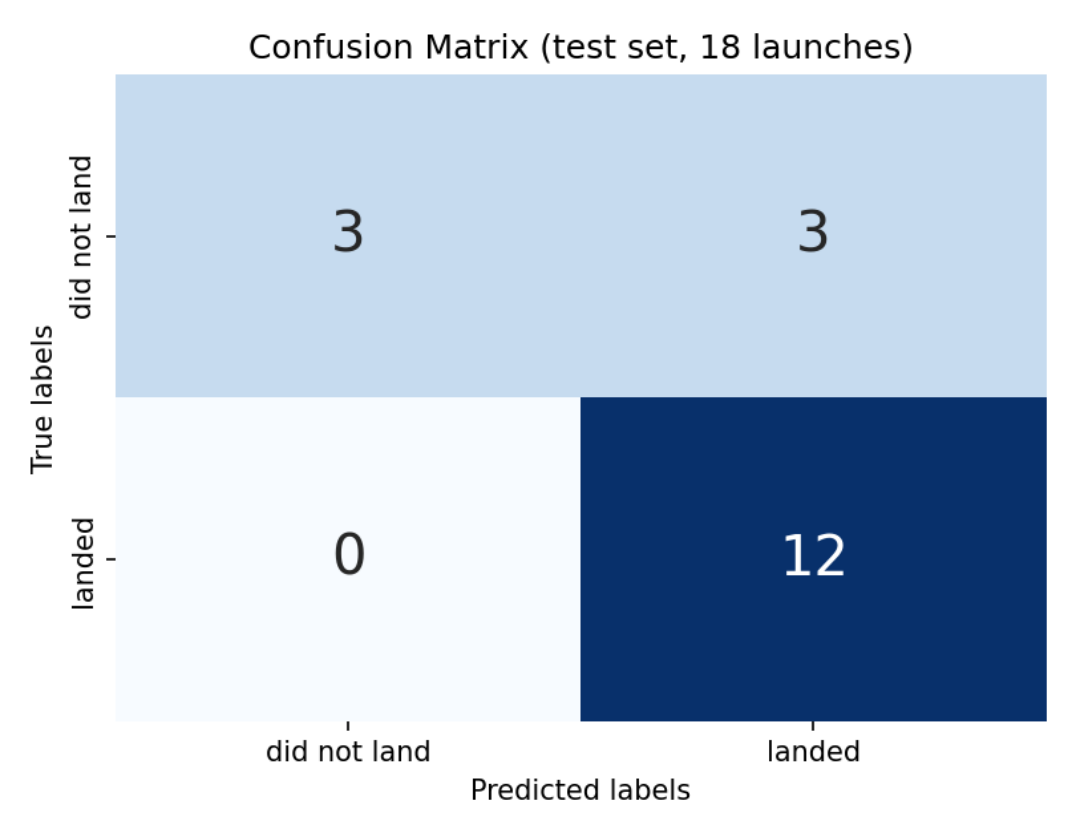


Classification Accuracy

- All four tuned models reach 83.3% test accuracy (15 of 18 correct)
- Best 10-fold CV accuracy: SVM and Decision Tree 84.8%, KNN 83.4%, Logistic Regression 82.1%
- With only 18 test launches the models cannot be separated on accuracy alone



Confusion Matrix



- All four models give the same confusion matrix on the 18 test launches
- 12 landings and 3 non-landings are predicted correctly
- The 3 errors are all false positives: launches that did not land but were predicted to land
- No landing was missed: recall 100%, precision for “landed” 80% (12 of 15)

Conclusions

- Landing success rose from 0% (up to 2013) to 84–90% in 2019–2020
- Success rises with flight number: 43% (flights 1–30), 73% (31–60), 83% (61–90)
- KSC LC-39A and VAFB SLC-4E landed about 77% of launches vs 60% at CCAFS SLC-40
- ES-L1, GEO, HEO and SSO reached 100% (1–5 launches each); VLEO 86%; GTO is lowest at 52%
- Payloads above 7,000 kg landed more often than lighter ones at all three sites
- All four classifiers tie at 83.3% test accuracy; the only errors are false positives (3 of 18)

Appendix

- Notebooks: data collection (API), web scraping, data wrangling, EDA with SQL, EDA with visualization, Folium launch-site map, machine learning prediction
- Plotly Dash app: spacex-dash-app.py
- Datasets: dataset_part_1/2/3.csv, spacex_web_scraped.csv, my_data1.db (SQLite)
- Next slide: model tuning details
- Github : https://github.com/harshit-09-ds/IBM-Data-Science-Capstone-SpaceX

Appendix: Model Tuning Details

Model	Best hyperparameters (GridSearchCV, cv = 10)	CV accuracy	Test accuracy
Logistic Regression	C=1, penalty=l2, solver=lbfgs	0.821	0.833
SVM	kernel=sigmoid, C=1, gamma=0.0316	0.848	0.833
Decision Tree	gini, max_depth=2, max_features=log2, splitter=best	0.848	0.833
KNN	n_neighbors=6, p=1, algorithm=auto	0.834	0.833

- Split: 80/20 with random_state = 2 (72 train, 18 test); features standardized with StandardScaler.
- Decision Tree has no fixed seed in the notebook, so its chosen parameters and CV score can vary slightly between runs.

Thank you!

